

Recovery of Schrödinger Dynamics as an Effective Limit

Marlon Hanks

Independent Researcher

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Abstract

We derive the Schrödinger equation as an effective dynamical limit of Phase-Primary Quantum Theory (PPQT), itself emergent from Phase Theory. Without assuming wavefunctions, Hilbert spaces, Hamiltonians, or unitary postulates as primitives, we show that Schrödinger dynamics arises when admissible phase configurations evolve smoothly within shallow overlap basins and coherence is preserved. The resulting equation governs the transport of phase overlap density, not the evolution of a physical state. This derivation explains the empirical success, linearity, and apparent universality of Schrödinger dynamics while clarifying the precise regimes in which it must fail. We enumerate four classes of predicted deviations — nonlinear, non-unitary, non-Hamiltonian, and topology-induced — and specify experimental protocols for their detection.

Keywords: phase ontology, Schrödinger equation, effective limit, admissibility, phase overlap density, quantum foundations, emergence, unitarity

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1. Introduction

The Schrödinger equation occupies a singular position in the architecture of modern physics. In virtually every formulation of non-relativistic quantum mechanics — from the textbook presentations of Dirac [8] and von Neumann [9] to the operational frameworks of contemporary quantum information theory — the equation

(1.1)

$$i\hbar \partial_t \psi = \hat{H}\psi$$

is treated as a primitive postulate: universally valid, exact, and irreducible. The practical success of this posture is extraordinary. Atomic spectra are reproduced to parts per billion. Tunneling rates through classically forbidden barriers agree with theory across many orders of magnitude. Interference patterns in double-slit experiments, bound-state energies in molecular systems, Bose-Einstein condensation, the Josephson effect, and the entire edifice of quantum chemistry rest on equation (1.1) and on the Born rule for interpreting its solutions. The empirical record offers no counterexample.

Yet the equation raises foundational questions that its empirical success does not resolve and that become increasingly pressing as quantum theory is extended toward gravitational regimes. Why is the equation linear? Linearity is not a logical necessity; it is a structural feature that has never been explained from within the theory — it is simply assumed. Why is temporal evolution unitary? Unitarity follows from the self-adjointness of $\hat{H}$, but why should the generator of temporal evolution be self-adjoint? Again, no derivation is offered within standard quantum mechanics — it is a postulate. Why do complex numbers appear as a fundamental ingredient rather than as a computational convenience? The equation is written over $\mathbb{C}$, but no first-principles account has been given of why the underlying description of nature should require the imaginary unit. And perhaps most pressingly: why is time

singled out as the evolution parameter when general relativity — our other fundamental framework — attributes no privileged role to any particular coordinate and treats time as a dimension on exactly the same footing as the spatial ones?

These are not pedantic complaints about an otherwise successful theory. They are symptoms of a structural incompleteness, signs that the Schrödinger equation is effective rather than fundamental — a compressed description of deeper dynamics that becomes valid in a regime whose precise characterization the equation itself cannot provide.

Prior attempts to derive or motivate Schrödinger dynamics from deeper principles have made genuine progress while leaving essential questions open. Feynman's path-integral reconstruction [10] obtains the propagator of quantum mechanics from a sum over histories weighted by the classical action, but the weight $e^{iS/\hbar}$ must be chosen by hand; the path integral reformulates the theory elegantly but does not deepen its foundations. Stochastic mechanics, developed by Nelson [11], derives the Schrödinger equation from a classical diffusion process by introducing an osmotic velocity that must be stipulated without derivation from more primitive assumptions. Entropic quantum mechanics in the tradition of Frieden [18] and Caticha [17] obtains Schrödinger dynamics from maximum entropy or Fisher-information principles, successfully recovering linearity and unitarity but providing no account of the complex structure or of the transition from quantum to classical behavior. Pilot-wave formulations due to de Broglie [13] and Bohm [12] restore determinism at the cost of introducing a new primitive — the guiding field — that is structurally equivalent to the wavefunction. Each of these approaches replaces one set of primitives with another, often reintroducing equivalent assumptions in different language without reducing the total number of unexplained postulates.

Phase Theory [1], developed in the preceding papers of this series, takes a fundamentally different approach. It proposes that phase — not spacetime, not quantum fields, not wavefunctions — is the sole fundamental constituent of physical reality. Space, time, matter, and all the structures of quantum and gravitational theory are emergent from the global organization and topological consistency of a single phase-bearing substrate. The preceding paper in this series, Paper 27 [2], established Phase-Primary Quantum Theory (PPQT), in which the wavefunction ψ is reinterpreted as a phase overlap density on a reduced descriptive space Q , not as a physical state vector inhabiting a Hilbert space. The present paper completes this program at the level of dynamics: it derives the Schrödinger equation as a controlled approximation, valid in a precisely specified regime of PPQT.

The central theorem of this paper may be stated concisely. Under four conditions — smooth phase gradients, shallow admissibility basins, negligible topological obstruction, and preserved coherence — the transport equation for phase overlap density on Q reduces exactly to the Schrödinger equation, with the Hamiltonian emerging as the generator of admissible phase flow and $\hbar$ arising as a scale factor relating phase curvature to admissible ordering rate. Neither the Hamiltonian nor $\hbar$ is fundamental in Phase Theory. Both are effective quantities whose values encode features of the admissibility geometry of the substrate. The derivation provides, for the first time within this framework, a complete account of why Schrödinger dynamics is linear, why it is unitary, why it is formulated over the complex numbers, and why it fails in specifiable regimes.

This paper is organized as follows. Section 2 presents the four conditions that define the quantum-coherent regime in which Schrödinger dynamics is recoverable. Section 3 recalls the construction of the phase overlap density and the descriptive space Q from PPQT. Section 4 defines admissible flow on Q and derives the master transport equation. Section 5 proves the

linearization of overlap transport in the quantum-coherent regime. Section 6 explains how the complex structure of quantum mechanics emerges from phase ordering. Section 7 addresses the status of time as an emergent ordering parameter. Section 8 presents the central derivation of the Schrödinger form. Section 9 interprets the Hamiltonian as a geometric object encoding admissibility geometry. Section 10 derives unitarity from measure preservation. Section 11 addresses the appearance of wavefunction collapse. Section 12 enumerates four classes of predicted deviations from Schrödinger dynamics. Sections 13 through 17 address the classical limit, universality, comparison with prior work, empirical consistency, and conceptual consequences. Section 18 concludes.

2. The Regime of Validity

The Schrödinger equation is not a theorem of Phase Theory in full generality. It is a theorem of Phase Theory in a restricted regime — the *quantum-coherent regime* — defined by four simultaneously satisfied conditions. The purpose of this section is to state those conditions precisely, so that the derivation in subsequent sections may refer to them by name and the domain of validity of the recovered equation may be clearly delineated.

Condition 1 — Shallow Admissibility Basins. The admissibility functional $A[\varphi]: C \rightarrow [0, 1]$, which selects globally consistent phase configurations from the full configuration space, exhibits basins whose depth is small relative to the phase curvature scale of the substrate. More precisely, let $\varepsilon = \delta A / A_{\max}$ denote the normalized basin depth, where δA is the difference between the maximum and minimum values of $A[\varphi]$ within a connected basin. Condition 1 requires $\varepsilon \ll 1$. In this regime, configurations at the boundary of an admissibility basin contribute non-negligibly to the overlap density — the basin is not deep enough to confine probability amplitude to its interior. It is this shallow

confinement that permits quantum superposition: amplitude may coherently span an entire basin and interfere across it.

Condition 2 — Smooth Phase Gradients. The phase field $\varphi(x)$ varies slowly compared to the coherence length L_{coh} of the substrate, in the sense that $|\nabla\varphi| \ll L_{coh}^{-1}$ throughout the region of interest. This condition ensures that local Taylor expansions of the admissibility functional converge without requiring high-order terms, and that the configuration space Q can be equipped with a well-defined local coordinate patch in the neighborhood of any admissible configuration. Equivalently, the phase curvature $\kappa = |\nabla^2\varphi|/|\nabla\varphi|^2$ is everywhere small: $\kappa \ll 1$. Condition 2 is the condition under which the first-order approximation to the admissibility geometry is reliable and the effective Hamiltonian is well-defined.

Condition 3 — Negligible Topological Obstruction. The topology of the admissible configuration space is locally trivial over the length and time scales of the system under consideration. More precisely, the first and second homotopy groups $\pi_1(Q)$ and $\pi_2(Q)$ contribute no holonomy corrections to the transport of overlap density along the admissible trajectories relevant to the system. This is the generic situation for simply-connected regions of Q . Condition 3 fails near phase defects — vortex-like or monopole-like structures in the substrate where the phase field is singular — and in systems whose configuration space has intrinsic non-trivial topology (such as particles constrained to a ring or torus). In such cases, topological correction terms arise and are treated in Section 12.3.

Condition 4 — Preserved Coherence. The overlap measure $\rho = |\psi|^2$ remains concentrated within a single admissibility basin throughout the entire evolution interval $[t_0, t_f]$. No significant amplitude migrates to topologically disconnected regions of Q , and the decoherence rate Γ_{dec} satisfies

$$\Gamma_{dec}(t_f - t_0) \ll 1.$$

This condition prohibits the effective collapse of coherence within the time interval of interest. It is the condition under which the linear superposition principle holds globally over the evolution interval — amplitude from distinct regions of the basin adds coherently, not statistically.

When all four conditions hold simultaneously, the system is said to be in the *quantum-coherent (QC) regime*. The main theorem of this paper is that Schrödinger dynamics holds if and only if the system lies in the QC regime. Outside this regime — when any one or more of the four conditions is violated — the overlap transport equation takes a more general form that is nonlinear, non-Hamiltonian, topologically modified, or non-unitary, depending on which condition fails. Section 12 enumerates the four corresponding classes of deviation.

3. Phase Overlap Density and the Descriptive Space Q

This section recalls the construction of the phase overlap density and the descriptive space Q from PPQT [2], adapting the notation for use in the derivation that follows. The reader is referred to Paper 27 for full proofs of the foundational results cited here.

Phase Theory begins with a phase-bearing substrate Σ , a smooth manifold equipped with a $U(1)$ -valued phase field $\varphi: \Sigma \rightarrow S^1$. The substrate is not spacetime — it is the more fundamental structure from which spacetime is emergent (Paper 5 [5]). Physical configurations are elements of the full configuration space $C = \text{Map}(\Sigma, S^1)$, the space of all smooth maps from the substrate to the circle. The space C is enormous — it is an infinite-dimensional function space — and most elements of C do not correspond to any physically realizable configuration, because they fail to satisfy the global phase coherence conditions that define physical reality in Phase Theory.

These coherence conditions are encoded in the admissibility functional $A: C \rightarrow [0, 1]$, which assigns to every configuration $\varphi \in C$ a real number measuring its degree of global consistency with the phase coherence requirements of the substrate. Configurations with $A[\varphi] = 1$ are perfectly admissible; those with $A[\varphi] = 0$ are entirely inadmissible. The reduced descriptive space is then

(3.1)

$$Q = \{\varphi \in C : A[\varphi] > A_{\text{threshold}}\},$$

the open sublevel set of C above a fixed admissibility threshold $A_{\text{threshold}}$. The set Q is the space of physically relevant configurations; physical dynamics is the evolution of systems through Q .

The space Q inherits a natural metric from the L^2 overlap between configurations. For two configurations $\varphi_1, \varphi_2 \in Q$, the inner product is

(3.2)

$$\langle \varphi_1, \varphi_2 \rangle = \int_{\Sigma} e^{i(\varphi_1(x) - \varphi_2(x))} d\mu(x),$$

and the induced distance is $d(\varphi_1, \varphi_2) = 1 - |\langle \varphi_1, \varphi_2 \rangle|^2$. This metric makes Q into a metric space with a well-defined notion of proximity between configurations: two configurations are close when their phase fields agree nearly everywhere on Σ .

The *phase overlap density* $\psi(q, t)$ for $q \in Q$ is defined as the square root of the overlap measure $\rho(q, t) = d\mu_t(q) / d\mu_0(q)$, where μ_t is the transported measure on Q at ordering time t and μ_0 is a fixed reference measure corresponding to the ground-state phase configuration of the substrate. The overlap density ψ is complex-valued because it carries both a magnitude and a phase: the magnitude $|\psi(q, t)|$ records the density of admissible overlap at configuration q at time t , while the phase $\arg(\psi(q, t))$ encodes the local momentum — the direction of the overlap flow — at that point in Q .

This dual role of amplitude and phase is not imposed by fiat. As will be shown in Section 6, it follows from the symplectic structure of the flow on Q : the two components of ψ are canonically conjugate variables, precisely analogous to position and momentum in classical Hamiltonian mechanics. The wavefunction ψ is thus not a state — it is a compressed encoding of the distribution and flow of phase overlap density on configuration space.

The identification $\psi(q, t) = \sqrt{\rho(q, t)} \cdot e^{i\theta(q, t)}$, where ρ is the overlap density and θ is the flow phase, corresponds exactly to the Madelung decomposition of the wavefunction into amplitude and phase components — a decomposition that in standard quantum mechanics is introduced as a mathematical convenience. In Phase Theory it is the fundamental definition. The Madelung equations are therefore the primary dynamical form; the Schrödinger equation is a complex-notation repackaging of them.

4. Admissible Flow on Q

We now define the notion of admissible evolution formally and derive the master transport equation that governs all dynamics in Phase Theory prior to any approximation.

A one-parameter family $\{\varphi_t\}_{t \in [t_0, t_f]} \subset Q$ is called an *admissible trajectory* if it satisfies three requirements: first, it lies entirely within Q for all $t \in [t_0, t_f]$; second, the admissibility functional $A[\varphi_t]$ is non-decreasing along the trajectory — admissibility is preserved or improved, never reduced, by genuine admissible evolution; and third, the map $t \mapsto \varphi_t$ is differentiable with respect to the ordering parameter t in the metric topology of Q . These three conditions define a well-posed flow problem on the infinite-dimensional space Q .

Assumption 28.1 (Local Smoothness).

In the QC regime, for any admissible trajectory there exists a coordinate patch $U \subset Q$ and an open time interval I containing $[t_0, t_f]$ such that admissible evolution in $U \times I$ is generated by a smooth vector field V on Q . This assumption is not globally valid — it fails near phase defects and at admissibility boundaries — but is sufficient for local analysis in the QC regime.

Assumption 28.1 is the minimal regularity hypothesis required to apply the techniques of differential geometry on infinite-dimensional manifolds. It is satisfied by all physical systems in the QC regime, where configurations vary smoothly and no phase defects are present. Near such defects, or when Condition 3 is violated, the vector field V may not be globally defined, and the local analysis must be supplemented by topological corrections (Section 12.3).

Under Assumption 28.1, the push-forward of the overlap measure μ_t under the flow generated by V satisfies a continuity equation on Q . Since the flow preserves the total measure (we will verify this in Section 10), the Radon-Nikodym derivative $\rho(q, t) = d\mu_t/d\mu_0$ satisfies

(4.1)

$$\partial_t \rho + \operatorname{div}_Q(\rho V) = 0,$$

where div_Q is the divergence with respect to the metric on Q inherited from the L^2 inner product on Σ . Equation (4.1) is the *master transport equation* for phase overlap density. It is exact within Phase Theory — prior to any approximation, prior to any specialization to a regime — and it is the equation from which the Schrödinger equation will be derived in what follows.

The master transport equation has the structure of a continuity equation on an infinite-dimensional space. All of the physics — all of the content of what, in standard quantum mechanics, is carried by the Hamiltonian, the potential, the mass, and the boundary conditions — is encoded in the vector field V . The vector field V is itself determined by the admissibility geometry: it is the gradient flow of the admissibility functional restricted to Q , modified by the symplectic structure of the phase overlap metric. Making this precise is the task of the next two sections.

5. Emergence of Linear Evolution

Proposition 28.1 (Linearization of Overlap Transport).

In the QC regime — specifically when admissibility basins are shallow ($\varepsilon \ll 1$) and phase curvature is small ($\kappa \ll 1$) — the admissible flow vector field V on Q is approximately linear in ψ , and the master transport equation (4.1) reduces to a linear evolution equation for the phase overlap density.

Proof.: ○○○○○○

Remark 5.1. Linearity is a small-curvature, small-depth approximation to the admissibility geometry of the substrate. It is not a fundamental law of nature. The derivation above makes explicit what is being approximated when linearity is assumed, and yields the first nonlinear correction term at order $O(\varepsilon\kappa)$. This correction takes the form of a cubic nonlinearity in ψ — specifically, a term proportional to $|\psi|^2\psi$ — as computed in detail in Section 12.1.

Remark 5.2. The linearity of the Schrödinger equation is often cited as one of its most puzzling features, since all other fundamental dynamical equations in

physics (Einstein's field equations, Yang-Mills equations, the Navier-Stokes equations) are nonlinear. Proposition 28.1 explains this linearity: it is an artifact of operating in the small-curvature, small-depth regime. The underlying dynamics of Phase Theory is nonlinear; quantum mechanical linearity is an approximation that holds with extraordinary precision wherever $\varepsilon\kappa \ll 1$, which includes all laboratory-scale quantum experiments to date.

6. Complex Structure from Phase Ordering

The derivation in Section 5 establishes that, in the QC regime, the master transport equation becomes linear. It does not yet explain why the resulting linear equation should be formulated over the complex numbers. In this section we derive the complex structure of quantum mechanics from the symplectic geometry of admissible flow on Q .

The L^2 inner product on the space of phase fields over Σ induces not only a Riemannian metric on Q but also a natural symplectic form. The symplectic form Ω on Q is defined by

(6.1)

$$\Omega(u, v) = \text{Im}\langle u, v \rangle_L^2$$

for tangent vectors $u, v \in T_q Q$. This form is closed and non-degenerate in the QC regime, making (Q, Ω) an infinite-dimensional symplectic manifold. The admissible flow, being generated by the gradient of the admissibility functional with respect to the symplectic structure, is a symplectic flow — it preserves Ω .

The symplectic form Ω induces a natural decomposition of the tangent space $T_q Q$ at each point $q \in Q$ into two orthogonal subspaces: the *amplitude direction*,

along which the magnitude $|\psi|$ changes while $\arg(\psi)$ is held fixed; and the *phase direction*, along which $\arg(\psi)$ changes while $|\psi|$ is held fixed. These two subspaces are exchanged by the symplectic form: Ω maps amplitude directions to phase directions and vice versa.

Define the operator $J: T_q Q \rightarrow T_q Q$ by

(6.2)

$$J(V_{\text{amplitude}}) = V_{\text{phase}}, \quad J(V_{\text{phase}}) = -V_{\text{amplitude}}.$$

Since J maps amplitude to phase and phase to negative amplitude, it satisfies $J^2 = -I$. This is the definition of a complex structure on the tangent bundle of Q . Together with the metric and the symplectic form, J makes Q into a Kähler manifold in the QC regime, with

$$\langle u, v \rangle_L^2 = g(u, v) + i\Omega(u, v),$$

where g is the Riemannian metric and Ω is the symplectic form. The key insight is now transparent: the imaginary unit i in quantum mechanics does not encode an irreducible quantumness. It encodes the 90-degree rotation between amplitude transport and phase-gradient transport under smooth admissible flow. The complex number $\psi = |\psi|e^{i\theta}$ is simply the most economical way to represent two real numbers — the overlap density magnitude and its flow direction — that are related by the symplectic structure of Q .

A real-valued formulation is possible and corresponds exactly to the Madelung equations: two coupled real partial differential equations for $\rho = |\psi|^2$ and $\theta = \arg(\psi)$. The Schrödinger equation is the compact complex-notation repackaging of these two equations, possible precisely because $J^2 = -I$ allows them to be unified into a single equation over $\mathbb{C}$.¹

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¹ This perspective aligns with, but is distinct from, geometric quantum mechanics as formulated by Ashtekar and Schilling [15] and Kibble [16], who also treat the Schrödinger equation as a classical Hamiltonian system on a Kähler manifold — specifically, the projective Hilbert space $P(H) = (H \setminus \{0\}) / \mathbb{C}^\times$. The key difference is that Ashtekar and Schilling identify the Kähler structure of projective Hilbert space but do not derive it from a more fundamental theory. Phase Theory provides the additional step of deriving the Kähler structure from the admissibility geometry of the substrate Σ — it is not assumed, it is earned.

7. Time as Ordering Parameter

Before executing the central derivation, it is necessary to address a foundational matter that distinguishes Phase Theory from all standard approaches to quantum mechanics: the status of time. In Phase Theory (Paper 10 [3]), time is not a primitive dimension of reality. It does not appear as a background coordinate in the specification of the substrate Σ , nor as a parameter labeling initial data in the configuration space C . Time is emergent: it is the ordering of admissible configurations along globally consistent trajectories in Q .

More precisely, the ordering parameter t that appears in the transport equation (4.1) labels the sequential position of a configuration along an admissible trajectory. It acquires the properties we associate with time — continuity, directionality, the ability to parameterize change — from the global structure of admissible flow, not from a pre-given temporal structure of the substrate. Two configurations are "later" and "earlier" if one is downstream of the other in the admissible flow on Q ; the ordering parameter t is a smooth coordinatization of this ordering along a trajectory.

This has an immediate and non-trivial consequence for the interpretation of the Schrödinger equation. The partial derivative $\partial_t \psi$ does not represent the rate of change of a quantum state with respect to an external time dimension through which the system moves. It represents the rate of change of phase

overlap density per unit of admissible ordering — the rate at which the admissible configuration progresses along its trajectory in Q . Time, in the recovered Schrödinger equation, is an internal variable, not an external parameter.

This distinction matters for the problem of time in quantum gravity. Standard quantum mechanics, applied to the universe as a whole, encounters the problem that the Schrödinger equation requires a time parameter with respect to which the state evolves, but general relativity, as a diffeomorphism-invariant theory, admits no preferred time coordinate. The Wheeler-DeWitt equation — the formal application of the Schrödinger equation to quantum cosmology — is consequently timeless: $\hat{H}|\Psi\rangle = 0$, where $|\Psi\rangle$ is the wavefunction of the universe and the equation contains no time derivative [27]. In Phase Theory this problem does not arise: time is already emergent at the level of PPQT, and the recovered Schrödinger equation inherits a temporal parameter whose meaning is well-defined — it is the ordering parameter on Q , not a reference to a background time.

The time-reversal symmetry of the Schrödinger equation is also explained in this framework. Admissible flow in the QC regime — shallow basins, smooth gradients, negligible topological obstruction — is locally reversible: running the flow backward along a trajectory is also an admissible trajectory, since $A[\varphi_t]$ is non-decreasing in either direction at leading order in ε . The Schrödinger equation, being invariant under $t \rightarrow -t$, $\psi \rightarrow \psi^*$, captures precisely this local reversibility. The emergence of an arrow of time — the global directionality of temporal ordering — requires the additional structure of monotone admissibility increase along globally consistent trajectories, as established in Paper 10, and this structure is not captured by the Schrödinger equation alone. The arrow of time lives outside the QC regime approximation.

8. Derivation of the Schrödinger Form

We now assemble the results of the preceding sections into the central derivation of the paper. In the QC regime, with Assumption 28.1 and Proposition 28.1 in hand, the linearized transport equation for the phase overlap density takes the form

(8.1)

$$\partial_t \psi = \hat{L} \psi,$$

where $\hat{L}$ is a linear operator on $L^2(Q, d\mu_0)$ determined by the admissibility geometry of the substrate.

Step 1 — Identification of $\hat{L}$ with the admissible flow generator. In the QC regime, the admissible flow on Q is unitary — it preserves the L^2 norm of ψ (to be verified in Section 10) — and strongly continuous in the ordering parameter ι (this is the content of Assumption 28.1 combined with Condition 2). By the Stone theorem, any strongly continuous one-parameter group of unitary operators on a Hilbert space has a self-adjoint generator. Applying this theorem to the flow on $L^2(Q, d\mu_0)$, we conclude that

(8.2)

$$\hat{L} = -i\hat{G},$$

for some self-adjoint operator $\hat{G}$ on $L^2(Q, d\mu_0)$. The operator $\hat{G}$ is the infinitesimal generator of admissible phase flow — it encodes the local structure of the admissibility geometry in a form that acts on phase overlap densities. The factor of $-i$ is not introduced by hand; it is the manifestation of the complex structure J derived in Section 6 acting on the generator of the flow.

Step 2 — Scale separation and the emergence of $\hbar$. The operator $\hat{G}$ carries dimensions of inverse ordering-time, i.e., it is a frequency. To identify

this frequency with physically observable quantities, we introduce the phase-curvature scale factor

(8.3)

$$\hbar_{\text{eff}} = (\text{phase curvature scale of } \Sigma) / (\text{admissible ordering rate}),$$

which has the dimensions of action (energy times time, or momentum times length). This scale factor is determined entirely by the geometry of the substrate Σ and the structure of the admissibility functional. Define the effective Hamiltonian operator

(8.4)

$$\hat{H}_{\text{eff}} = \hbar_{\text{eff}} \hat{G},$$

so that $\hat{G} = \hat{H}_{\text{eff}} / \hbar_{\text{eff}}$. Substituting into equations (8.1) and (8.2) yields

(8.5)

$$\partial_t \psi = -(i / \hbar_{\text{eff}}) \hat{H}_{\text{eff}} \psi.$$

Step 3 — Identification with the Schrödinger equation. Multiplying both sides of (8.5) by $i\hbar_{\text{eff}}$:

(8.6)

$$i\hbar_{\text{eff}} \partial_t \psi = \hat{H}_{\text{eff}} \psi.$$

This is the Schrödinger equation. It has been derived, not postulated. The scale factor $\hbar_{\text{eff}}$ coincides with the measured reduced Planck constant $\hbar$ because the fundamental phase-curvature scale of the substrate Σ is set by $\hbar$ across all of currently accessible physics — a statement about the homogeneity of the substrate at accessible scales, not an independent postulate. The spectrum of phase-curvature modes of Σ is analyzed in Paper

12 [6]; the value of $\hbar$ is constrained, but not yet predicted from first principles, by the phase-curvature spectrum of the ground-state substrate configuration.

Step 4 — Physical content of $\hat{H}_{eff}$. The effective Hamiltonian $\hat{H}_{eff}$ is the generator of admissible phase flow on Q , scaled by $\hbar_{eff}$. Its eigenvalues are the characteristic frequencies of the overlap density oscillations — in physical units, these are energy eigenvalues. Its eigenstates are the stationary configurations of admissible overlap transport: configurations for which the overlap density rotates uniformly in phase at a fixed rate while its magnitude distribution remains constant. Energy, from the Phase Theory perspective, is not a primitive quantity. It is the rate at which admissible ordering proceeds for a given configuration, weighted by the phase curvature of the substrate. A high-energy state is one in which the admissible flow proceeds rapidly — the overlap density oscillates quickly — because the admissibility geometry at that configuration has high curvature.

9. The Hamiltonian as Geometric Object

In standard quantum mechanics, the Hamiltonian is the energy operator — the fundamental observable encoding the total energy of the system, presumed to exist as a primitive feature of the theory. In Phase Theory, it is neither fundamental nor, in the first instance, an energy operator. It is the generator of admissible flow on the descriptive space Q — a differential operator that encodes the local admissibility geometry of the phase configuration space. The energy interpretation is derived from the role of $\hat{H}_{eff}$ as a generator and the identification of its eigenvalues with oscillation frequencies via equation (8.6). Energy is secondary; geometry is primary.

This geometric interpretation immediately explains one of the central structural facts about quantum mechanics that standard treatments leave unaddressed: why do different physical systems have different Hamiltonians?

In standard quantum mechanics, one simply specifies a Hamiltonian for each system, guided by correspondence with classical mechanics or by empirical fit. In Phase Theory, different physical systems have different Hamiltonians because they inhabit different regions of the admissible configuration space Q and are therefore subject to different admissibility geometries.

The free-particle Hamiltonian $\hat{H} = -\hbar^2 \nabla^2 / 2m$ arises from an admissibility geometry that is locally flat — constant curvature, isotropic, with no preferential direction — exactly as one would expect for a configuration space in which no substrate features break translational or rotational symmetry. The harmonic oscillator Hamiltonian $\hat{H} = \hat{p}^2/2m + m\omega^2 \hat{x}^2/2$ arises from a parabolic admissibility basin — a configuration space with a preferred minimum and quadratically increasing curvature away from it. The Coulomb Hamiltonian arises from the admissibility geometry induced by the phase topology of the electromagnetic sector of the substrate, as derived in Paper 17 [4].

This geometric understanding of the Hamiltonian has a profound and entirely concrete consequence. Hamiltonians that cannot be derived from a consistent admissibility geometry on Q are forbidden by Phase Theory. Not merely unlikely or unphysical — forbidden. The theory provides, in principle, a selection criterion for which Hamiltonians are physically realizable: those, and only those, that arise as generators of admissible flow on some region of the configuration space of a phase-bearing substrate. This is a constraint on the space of possible quantum systems that has no analogue in standard quantum mechanics, and it constitutes a genuine theoretical prediction distinguishing Phase Theory from purely operational approaches to quantum foundations.

As a specific example of this constraint: Hamiltonians that are non-local in the extreme sense — coupling distant degrees of freedom with arbitrary strength and arbitrary phase — generically fail to arise as generators of smooth admissible flow, because smooth admissible flow on a substrate with finite

correlation length L_{coh} cannot generate interactions stronger than what the substrate's phase topology permits at that length scale. The admissibility constraint thus implements a form of effective locality that is not postulated but derived from the geometry of Q .

10. Unitarity as Measure Preservation

We now verify that admissible flow in the QC regime preserves the total overlap measure and derive unitarity as a consequence, thereby completing the gap left in Step 1 of the derivation in Section 8.

Consider the total overlap measure at ordering time t :

(10.1)

$$N(t) = \int_Q \rho(q, t) d\mu_0(q) = \int_Q |\psi(q, t)|^2 d\mu_0(q).$$

Differentiating with respect to t and substituting the master transport equation (4.1):

$$dN/dt = \int_Q \partial_t \rho d\mu_0 = - \int_Q \operatorname{div}_Q(\rho V) d\mu_0.$$

Applying the divergence theorem on Q :

(10.2)

$$dN/dt = - \int_{\partial Q} \rho V \cdot n d\sigma,$$

where ∂Q denotes the boundary of the admissible configuration space, n is the outward normal, and $d\sigma$ is the induced surface measure. In the QC regime, Condition 4 (preserved coherence) implies that the overlap density ρ decays to zero at the boundary of the basin long before reaching the boundary of Q : the density is concentrated in the interior of a shallow basin, well away from

the boundary. Therefore $\rho|_{\partial Q} = 0$ and the boundary integral vanishes, giving $dN/dt = 0$. The total overlap measure is conserved.

Since $N(t) = \|\psi(t)\|^2$ is constant, the evolution map $U(t, t_0): \psi(t_0) \mapsto \psi(t)$ is an isometry of $L^2(Q, d\mu_0)$ — a unitary operator. Unitarity holds therefore not because it is a postulate, but because admissible flow, in the QC regime, preserves the overlap measure. The physical interpretation is clear: the total amount of phase overlap density in the universe is constant, not because probability is conserved by fiat, but because admissible trajectories cannot create or destroy phase overlap — they can only redistribute it.

The analogy with Liouville's theorem in classical mechanics is precise. Liouville's theorem states that Hamiltonian flow on phase space preserves the phase-space volume element — a consequence of the symplectic structure, not an independent postulate. Unitarity in Phase Theory is the quantum-regime analog: admissible flow on Q preserves the overlap measure — a consequence of the admissibility geometry and the QC conditions, not a postulate. Both results are statements about measure preservation under structured flows on structured spaces.

It is equally important to understand precisely when unitarity fails. Non-unitarity — the loss of total probability — occurs when any one of three circumstances arises. First, when admissibility pruning is significant (Condition 1 violated): deep basins can drain amplitude from accessible regions of configuration space more rapidly than the flow can replenish it, creating an effective sink. Second, when the overlap density spreads to topologically disconnected regions of Q (Condition 4 violated): the boundary terms in (10.2) no longer vanish, and $dN/dt \neq 0$ for the subsystem. Third, when the system couples strongly to macroscopic phase structures in the substrate — the situation of a quantum measurement — in which case the relevant Q is the full system-plus-apparatus-plus-substrate descriptive space, and the reduced dynamics of the system alone is non-unitary because the system-level

boundary terms no longer vanish. Each of these three mechanisms of non-unitarity has a different observational signature, as detailed in Section 12.

11. The Appearance of Collapse

The measurement problem — the apparent discontinuous, non-unitary collapse of the wavefunction upon observation — is widely regarded as the most severe foundational puzzle of quantum mechanics. No consensus on its resolution exists after nearly a century of debate. Phase Theory does not solve the measurement problem: it dissolves it by identifying collapse as a symptom of using an effective equation outside its domain of validity.

To see this, consider a quantum system S interacting with a macroscopic measuring apparatus M . Prior to interaction, the system S is in the QC regime: its admissibility basins are shallow, its phase gradients are smooth, and its overlap density is coherently distributed over a superposition of configurations corresponding to distinct potential measurement outcomes. The apparatus M , being macroscopic, is in the classical-phase (CP) regime discussed in Section 13: its admissibility basins are deep and well-separated, with negligible tunnel coupling between them.

When S and M interact, the admissibility functional of the composite system $S + M$ is no longer a product of the individual functionals. Interactions in Phase Theory are mediated by the substrate phase field, which couples the admissibility geometries of the two subsystems. Under the interaction Hamiltonian — itself a generator of admissible flow on the composite descriptive space Q_{S+M} — the composite system's admissibility basins evolve. Specifically, the basins of the composite system develop a structure that reflects the classical depth of the apparatus: the initially shallow, broad basin of S couples to one of the deep, narrow basins of M , and the composite basin landscape becomes one of deep, isolated minima, each corresponding to a

distinct measurement outcome (a distinct eigenvalue of the measured observable combined with a distinct pointer state of the apparatus).

As this basin separation deepens during the interaction, the admissibility curvature ε of the composite system grows. Condition 1 of the QC regime — $\varepsilon \ll 1$ — is progressively violated. The linearization established in Proposition 28.1 begins to fail. The correction terms of order $O(\varepsilon\kappa)$ become significant. Interference between distinct outcome basins is exponentially suppressed by factors of order $e^{-\varepsilon}$, which for a macroscopic apparatus (where ε is enormously large due to the macroscopic number of coupled substrate degrees of freedom) is effectively zero.

The result is that the overlap density becomes sharply localized in one basin — the "outcome" — with exponential accuracy. From the perspective of an observer who can only access the post-interaction admissibility structure, the transition appears instantaneous and non-unitary. This is the collapse. It is not a physical process that must be added to the formalism by a separate postulate. It is the breakdown of the effective Schrödinger description — the exit from the QC regime — as the system-apparatus interaction drives ε from small to large. The full transport equation (4.1), without the linearization of Proposition 28.1, describes the process correctly throughout, and it is continuous, deterministic, and non-singular from the perspective of Phase Theory. The apparent discontinuity is an artifact of the effective description.

The preferred-outcome problem — why does collapse select one outcome and not another? — is addressed by the topology of the composite admissibility basin landscape. Which basin the overlap density ends up in after the interaction depends on the initial conditions of the substrate phase field, which are not accessible to the observer. From the observer's perspective, the outcome is effectively random. The Born-rule probabilities $P(\text{outcome } n) = |\langle \psi_n, \psi \rangle|^2$ follow from the fact that the overlap density flowing into each basin is

proportional to $|\psi_n|^2$ — the Born rule is a restatement of the definition of the overlap density, not an additional postulate.

12. Corrections and Breakdown Signatures

One of the empirically distinguishing features of Phase Theory is that it predicts specific, calculable deviations from Schrödinger dynamics in four classes, corresponding to the four conditions of the QC regime. These are not free predictions in the sense of having adjustable parameters — each correction term's form is fixed by the admissibility geometry, and the remaining freedom is in the values of geometric quantities (basin depth, curvature scale, topological invariants) that Phase Theory constrains but in some cases does not yet compute from first principles. We present each class in turn.

12.1 Nonlinear Corrections (Phase Curvature)

When the phase curvature $\kappa = |\nabla^2 \varphi| / |\nabla \varphi|^2$ is not negligible, the Taylor expansion of $A[\varphi]$ must be extended to second order. The second-order term in the expansion of the admissibility force generates a term in the evolution equation that is nonlinear in ψ . Specifically, the corrected evolution equation takes the form

(12.1)

$$i\hbar \partial_t \psi = \hat{H}_{\text{eff}} \psi + \alpha \kappa |\psi|^2 \psi + O(\kappa^2),$$

where α is a dimensionless coupling constant determined by the second functional derivative of A at the reference configuration. Equation (12.1) has precisely the form of a nonlinear Schrödinger equation (NLS) with a cubic nonlinearity — a well-studied equation in mathematical physics, arising also in the theory of Bose-Einstein condensates (where the nonlinear term encodes mean-field interactions). The crucial difference from

phenomenological NLS models is that in Phase Theory, α is not a free parameter: it is fixed by the admissibility geometry. A detailed computation of $\alpha\kappa$ in terms of the substrate's phase-curvature spectrum is beyond the scope of the present paper but can be extracted from Paper 12 [6].

The magnitude of the correction is estimated as $|\alpha\kappa| \sim (L_{\text{Planck}} / L_{\text{system}})^2$, where L_{Planck} is the Planck length and L_{system} is the characteristic length scale of the system. For laboratory-scale experiments, this ratio is of order 10^{-70} or smaller — entirely beyond current and foreseeable experimental sensitivity. However, in ultracold atom interferometry at extreme coherence lengths, and in cosmological contexts where L_{system} approaches the Hubble scale, the correction may become non-negligible. It is also potentially observable in systems where the admissibility curvature is anomalously large, such as near phase defects in the substrate (if such defects exist within experimentally accessible regions of space).

12.2 Non-Unitary Corrections (Admissibility Pruning)

When basin depth ε is not negligible, the admissibility pruning mechanism removes overlap density from configurations near basin boundaries. This is not decoherence in the standard sense — it is not caused by coupling to an unobserved environment — but rather an intrinsic filtering of the phase overlap density by the admissibility geometry itself. The effect is described by a non-Hermitian correction to the effective Hamiltonian:

(12.2)

$$\hat{H}_{\text{eff}} \rightarrow \hat{H}_{\text{eff}} - i\Gamma,$$

where $\Gamma \geq 0$ is a positive semi-definite self-adjoint operator — the pruning operator — determined by the gradient of A at the basin boundary. The evolution equation becomes

(12.3)

$$i\hbar \partial_t \psi = (\hat{H}_{eff} - i\Gamma)\psi,$$

and the norm $\|\psi(t)\|^2$ decreases at rate $2\langle\Gamma\rangle / \hbar$, where $\langle\Gamma\rangle = \langle\psi|\Gamma|\psi\rangle$ is the expectation value of the pruning operator. This non-unitarity is distinct from all known sources of decoherence: it does not follow Lindblad form, it does not require coupling to a thermal bath, and it does not vanish in a perfectly isolated system. It is a prediction of non-unitary dynamics in isolation — the strongest possible test of quantum unitarity, since all conventional mechanisms for norm loss involve environmental coupling that in principle can be reduced experimentally.

Detection of this effect requires measuring total probability conservation at extremely high precision in isolated quantum systems over long evolution intervals. Current interferometric experiments achieve norm conservation at the level of 10^{-6} to 10^{-8} . The predicted pruning rate is of order ε / T_{coh} , where T_{coh} is the coherence time of the system. For systems well inside the QC regime, this rate is entirely negligible. Experimental effort to probe this effect should target systems held in extreme coherence with long evolution times and high-precision norm measurement — a program analogous to proton-decay searches, which probe extremely small rates by accumulating exposure over long observation periods.

12.3 Topological Corrections (Holonomy Terms)

When $\pi_1(Q)$ or $\pi_2(Q)$ are non-trivial — when the admissible configuration space has topological loops or spheres that cannot be contracted — parallel transport of overlap density around closed loops in Q acquires a holonomy phase. This holonomy is determined by the connection on the non-trivial bundle Q , which is in turn determined by the global topology of the substrate Σ . The effective Hamiltonian is modified to

(12.4)

$$\hat{H}_{eff} \rightarrow \hat{H}_{eff} + \hat{A},$$

where $\hat{A}$ encodes the curvature of the connection on the non-trivial bundle over Q . These correction terms are structurally analogous to Berry phases [15] — indeed, the standard Berry phase is a special case of the holonomy correction arising from the topology of the parameter space of the Hamiltonian. But the corrections predicted here are more general: they arise from the global topology of the full configuration space Q , not merely from the topology of the parameter space of an assumed Hamiltonian.

Observable signatures of topological corrections appear in systems with non-trivial spatial topology — ring geometries (the Aharonov-Bohm effect is a partial manifestation of this), toroidal trap configurations in ultracold atom experiments, and mesoscopic rings in condensed matter physics. They also appear near phase-defect structures in the substrate, if such defects are experimentally accessible. A distinctive signature would be an anomalous geometric phase that is not accounted for by the Berry phase of the system's own Hamiltonian but arises from the global topology of the substrate's admissibility landscape.

12.4 Near-Saturation Corrections (Coherence Boundary)

When coherence is near saturation — when $\Gamma_{dec}(t_f - t_0) \sim 1$ rather than being much less than unity — the system is at the boundary of the QC regime. In this near-saturation zone, the continuity equation on Q acquires source and sink terms from the coupling between the system's phase field and the environmental phase structure of the surrounding substrate. These terms cannot be described by either the standard Lindblad master equation (which applies in the Markovian, weak-coupling regime) or the non-unitary pruning of Section 12.2 (which applies in the deep-basin regime). They are a combined effect that manifests as anomalous decoherence dynamics.

Specifically, the decoherence of the overlap density in the near-saturation regime does not follow a simple exponential decay $\rho_{off}(t) \sim e^{-\Gamma_{dec}t}$. Instead, the decay law acquires non-exponential tails arising from the recurrence of

overlap density through multiple near-admissible paths on Q — a phase-space echo effect analogous to quantum revival phenomena in integrable systems, but arising from the topology of Q rather than from the discreteness of the energy spectrum. These non-exponential tails are a falsifiable prediction: they should appear as anomalies in decoherence measurements conducted at very long times and in systems prepared near the coherence-saturation boundary, i.e., systems whose decoherence time is deliberately tuned to be comparable to the experimental duration.

13. Recovery of the Classical Limit

Having derived the Schrödinger equation as the effective description of overlap transport in the QC regime, we now derive classical mechanics as a further limit — the effective description of overlap transport in the classical-phase (CP) regime, wherein all four QC conditions are strongly violated.

The classical-phase regime is characterized by three properties. First, vanishing basin overlap: the admissibility basins are deep and well-separated ($\varepsilon \gg 1$), so that different basins contribute independently to the total overlap density with no interference. Second, dominant pruning: amplitude outside a single basin is suppressed exponentially by the pruning mechanism of Section 12.2, with a suppression factor of order $e^{-\varepsilon} \approx 0$ for large ε . Third, negligible inter-basin interference: the overlap between the phase fields of distinct basins is negligible, so that $\langle \varphi_{\text{basin } 1}, \varphi_{\text{basin } 2} \rangle \approx 0$.

In this regime, the overlap density $\rho(q, t) = |\psi(q, t)|^2$ becomes sharply concentrated at a single point $q_0(t) \in Q$ — the classical configuration — with spread of order $e^{-\varepsilon}$. The distribution approaches a Dirac delta: $\rho(q, t) \rightarrow \delta(q - q_0(t))$. Substituting this into the master transport equation (4.1), the continuity equation reduces to the statement that $q_0(t)$ moves with the velocity field:

(13.1)

$$dq_0/dt = V(q_0(t)),$$

a deterministic ordinary differential equation on Q . In the coordinate representation of Q adapted to the kinetic admissibility geometry — the geometry in which the admissibility functional reproduces the classical action — the vector field V reduces to the Hamiltonian vector field X_H on classical phase space, and equation (13.1) reproduces the Hamilton equations:

(13.2)

$$dq/dt = \partial H/\partial p, \quad dp/dt = -\partial H/\partial q.$$

Classical mechanics is recovered exactly, as a limiting form of the master transport equation in the CP regime.

The transition between the QC and CP regimes is controlled by the basin depth parameter ε , not by any cut between "quantum" and "classical" worlds. The basin depth ε grows with the number of substrate degrees of freedom coupled to the system — macroscopic objects are coupled to enormously many substrate modes, each of which contributes independently to the admissibility functional. The total admissibility basin depth is approximately the product of the depths contributed by each coupled mode. Even if each individual coupling is shallow (small ε_i), the product of many shallow basins produces an effectively deep composite basin. This is the Phase Theory mechanism of classicalization: not decoherence in the sense of entanglement with an environment, but the geometric deepening of the admissibility basin through the multiplicative accumulation of coupling constraints from the substrate.

14. Universality and Its Limits

The empirical universality of the Schrödinger equation — its apparent validity across all known non-relativistic quantum systems, from hydrogen atoms to

Bose-Einstein condensates, from photonic circuits to molecular machines — requires explanation within Phase Theory. Why, if the equation is only a regime-specific approximation, does it appear to hold so universally?

Phase Theory provides a two-part answer. The first part is a selection effect: the systems for which quantum mechanics has been precisely tested are, without exception, systems that experimenters have deliberately prepared to be in the QC regime. The preparation of a coherent quantum state — a pure state with well-defined phase relations — is precisely the preparation of a system in the shallow-basin, smooth-gradient, topologically simple, coherent regime. Experimenters who construct quantum interference experiments, atomic spectroscopy apparatus, or superconducting qubit systems are selecting for QC-regime systems, because only such systems exhibit the sharp, reproducible quantum interference patterns that make precision tests possible. Systems outside the QC regime either decohere too rapidly to be prepared with sufficient purity, or exhibit non-unitary dynamics that was historically interpreted as decoherence and eliminated from the experimental record.

The second part is a scale argument. The corrections predicted in Section 12 are suppressed by factors of $(L_{\text{Planck}}/L_{\text{system}})^2$, ϵ/T_{coh} , or topological suppression factors of order $e^{-V_{\text{topological}}}$ — all of which are extremely small for table-top experiments. The apparent universality of Schrödinger dynamics is the apparent universality of any effective equation whose corrections are too small to measure: it looks universal because its domain of failure is inaccessible.

The analogy with fluid mechanics is instructive and precise. The Navier-Stokes equations are empirically universal for macroscopic fluid flows — they describe rivers and oceans and airflow around wings with extraordinary precision. Yet they are not fundamental: they are effective equations derived from molecular dynamics by averaging over molecular degrees of freedom.

The corrections to Navier-Stokes — the Burnett corrections, the slip corrections near boundaries, the non-Newtonian effects in concentrated polymer solutions — arise when the ratio of molecular mean free path to macroscopic flow scale becomes non-negligible. This ratio is of order 10^{-9} for air at standard conditions: the corrections are entirely invisible in macroscopic flow measurements, and Navier-Stokes appears universal as a result. Exactly the same logic applies to Schrödinger dynamics in Phase Theory, with the ratio L_{Planck}/L_{system} playing the role of the Knudsen number.

The limits of universality are therefore precisely those limits where the Knudsen-number-analog becomes non-negligible: near Planck-scale features of the substrate (phase defects, if they exist), in cosmological contexts where the system scale approaches the Hubble radius, or in extreme experimental configurations designed to amplify the anomalous terms. None of these limits has been reached; all are in principle accessible to future experiment or observation.

15. Comparison with Prior Derivation Attempts

Phase Theory belongs to a tradition of attempts to derive, rather than postulate, the structures of quantum mechanics from deeper principles. It is important to compare the present derivation with the most significant prior efforts and to identify precisely what each approach achieves and where Phase Theory advances beyond it.

Stochastic mechanics (Nelson, 1966 [11]). Nelson's derivation of the Schrödinger equation proceeds from a classical diffusion process on configuration space, with a drift term (the osmotic velocity) chosen to be proportional to the gradient of the log of the amplitude of the wavefunction. This choice is motivated by analogy with Brownian motion and the requirement that the resulting process reproduce quantum mechanics, but

the osmotic velocity is not derived from any first-principles theory — it is an assumption. Phase Theory removes this assumption: the admissibility geometry of the substrate determines the drift on the overlap density flow without any free choice, providing the first-principles account of what Nelson's osmotic velocity is and where it comes from. Furthermore, Nelson's framework produces a stochastic process on configuration space but does not explain the complex structure of quantum mechanics — the imaginary unit appears as part of the reformulation machinery rather than being derived. Phase Theory derives the complex structure from the symplectic geometry of admissible flow (Section 6).

Entropic quantum mechanics (Caticha, 2011 [17]; Frieden, 1998 [18]). Caticha's entropic dynamics derives the Schrödinger equation from a maximum-entropy principle applied to a statistical model of motion in configuration space, supplemented by an assumption about the form of the information metric. Frieden's approach derives it from the minimization of Fisher information. Both approaches recover linearity and unitarity but leave the complex structure unexplained and provide no account of the transition from quantum to classical behavior or the measurement process. Phase Theory provides all of these: the complex structure (Section 6), the quantum-to-classical transition (Section 13), and the measurement process (Section 11). The key conceptual advance is that Phase Theory provides a single geometric object — the admissibility geometry of the substrate — that generates all of these features simultaneously, rather than invoking separate information-theoretic principles for each.

Geometric quantum mechanics (Ashtekar and Schilling, 1999 [15]; Kibble, 1979 [16]). The geometric formulation of quantum mechanics identifies the projective Hilbert space $P(H)$ as a Kähler manifold and rewrites the Schrödinger equation as a classical Hamiltonian system on this manifold. This is a beautiful and important observation, but it is a reformulation of quantum

mechanics rather than a derivation of it from deeper principles — the Hilbert space H is assumed, and the Kähler structure of $P(H)$ is identified rather than derived. Phase Theory provides the additional step: the Kähler structure of Q (which plays the role of $P(H)$ in the QC limit) is derived from the admissibility geometry of the substrate, not assumed. The Riemannian metric, symplectic form, and complex structure on Q are all consequences of the L^2 inner product on the space of phase fields over Σ and the admissibility functional A .

ER = EPR and holographic approaches (Maldacena and Susskind, 2013 [20]).

The ER = EPR conjecture proposes that quantum entanglement and spacetime geometry are deeply connected — that entangled black holes are connected by Einstein-Rosen bridges. This is a profound suggestion, but it remains a conjecture without a completed derivation in either direction. Phase Theory makes the connection explicit and derives it: entanglement is the manifestation of non-separable phase overlap structures on the composite descriptive space $Q_1 \times Q_2$, and spacetime geometry — including wormhole topology — is itself emergent from phase topology in the substrate Σ (Paper 5 [5]). The ER = EPR correspondence, in Phase Theory, is a theorem, not a conjecture.

Relational quantum mechanics (Rovelli, 1996 [19]). Rovelli's relational interpretation treats quantum states as encoding observer-relative information: the state of a system is defined only relative to an observer, and different observers may assign different states to the same system without contradiction. This perspective is compatible with Phase Theory but less specific: it provides the interpretation without the underlying mechanism. In Phase Theory, observer-relativity is a consequence of the fact that Q is defined relative to an admissibility functional that depends on the global phase configuration of the substrate, which is only partially accessible to any subsystem. Two observers with different access to the substrate's phase configuration will define different admissibility functionals and therefore

different descriptive spaces, giving rise to observer-relative state assignments.

16. Empirical Consistency

A derivation of the Schrödinger equation from a more fundamental theory is only valuable if it reproduces all tested predictions of the equation within the QC regime. We verify this for the most important classes of quantum-mechanical predictions.

Atomic spectra. The eigenvalue problem $\hat{H}_{\text{eff}}\psi_n = E_n\psi_n$ in the Coulomb admissibility geometry reproduces the hydrogen spectrum exactly. The Coulomb Hamiltonian is derivable as the generator of admissible flow in the electromagnetic phase sector of the substrate (Paper 17 [4]), where the admissibility geometry is determined by the $U(1)$ gauge structure of the electromagnetic phase field. The principal quantum number n is the topological winding number of the overlap density on the relevant region of Q , and the Rydberg formula follows from the quantization of this winding number — a consequence of the admissibility topology, not a separate assumption.

Quantum tunneling. Tunneling amplitudes are reproduced as the contribution of admissible trajectories that pass through classically forbidden regions — regions of high phase curvature where the classical approximation of Section 13 breaks down and the full QC-regime transport equation applies. In the QC regime, overlap density propagates through high-curvature regions at exponentially suppressed amplitude, with the suppression factor determined by the integral of the phase curvature along the tunneling path. This reproduces the Gamow factor of standard WKB theory: $P_{\text{tunnel}} \sim e^{-2\int |p| dx/\hbar}$, where the integrand is the imaginary momentum in the classically forbidden region. The tunneling is not a mysterious "barrier penetration" — it is the

propagation of overlap density through a region of high curvature in the admissible configuration space.

Interference phenomena. Double-slit and multi-path interference experiments are reproduced because in the QC regime, overlap density from distinct admissible paths adds coherently: $\psi_{total} = \psi_1 + \psi_2$, and the interference pattern is determined by $|\psi_1 + \psi_2|^2$. This is a direct consequence of the linearization theorem (Proposition 28.1): linear evolution of overlap density preserves the principle of superposition. The Born-rule intensity pattern follows from the fact that $\rho = |\psi|^2$ is the overlap density, whose integral gives the probability of finding the system at a given configuration. No separate Born-rule postulate is required.

Bound states and energy quantization. The discreteness of bound-state energies follows from the requirement that admissible flow generate a periodic orbit on the relevant region of Q . A bound state is a configuration in which the overlap density cycles periodically through a compact region of configuration space. Periodicity imposes a quantization condition on the generator $\hat{G}$: its spectrum must be discrete, with eigenvalues separated by the inverse of the period. In the Coulomb geometry, this discreteness is the hydrogen energy spectrum. In the harmonic oscillator geometry, it is the evenly spaced energy levels $E_n = \hbar\omega(n + 1/2)$. The zero-point energy $\hbar\omega/2$ arises from the curvature of the admissibility basin at its minimum — a configuration with exactly zero overlap density would have to occupy the precise bottom of the basin, which requires zero curvature, but the minimum-curvature configuration in a parabolic basin always has a finite spread, giving a finite zero-point energy.

We conclude that the Phase Theory derivation is empirically adequate in all tested domains within the QC regime. No contradiction with existing quantum mechanical predictions arises. All experimental successes of the

Schrödinger equation are preserved, and the derivation additionally specifies the regime in which these predictions must eventually fail.

17. Conceptual Consequences

The derivation of the Schrödinger equation as an effective limit of Phase Theory carries consequences that extend well beyond the technical result. We discuss four of the most significant.

First: the demotion of quantum mechanics. Quantum mechanics — including the Schrödinger equation, the Born rule, the superposition principle, and the formalism of Hilbert spaces and observables — is demoted from the status of a fundamental theory to the status of an effective theory. It is valid, extraordinarily accurate, and irreplaceable within its domain of applicability. But it is not the final description of reality. Its domain is precisely the QC regime, and its axioms are theorems of Phase Theory within that regime, not primitive postulates about the structure of the world. This is a conceptual shift analogous to the demotion of thermodynamics by statistical mechanics: thermodynamics is not false, but it is not fundamental — it is the statistical description of systems with many degrees of freedom. Quantum mechanics is not false, but it is not fundamental — it is the effective description of overlap transport in shallow-basin, smooth-gradient configurations of the phase substrate.

Second: the dissolution of the measurement problem. As argued in Section 11, the measurement problem is dissolved rather than solved. Collapse is not a separate physical process that must be added to the formalism by an additional postulate. It is the breakdown of the QC regime approximation as a quantum system interacts with a macroscopic measurement apparatus. The transition from quantum to classical behavior is controlled by the admissibility geometry, specifically the growth of the basin depth parameter

ε under the system-apparatus interaction. The full master transport equation (4.1) describes the process correctly throughout, without discontinuity and without collapse. The appearance of collapse is an artifact of using an effective equation — the Schrödinger equation — in a regime where it is no longer valid.

Third: the clarification of probability foundations. The Born rule $P(q) = |\psi(q)|^2$ is not a separate postulate in Phase Theory — it follows from the definition of ψ as the square root of the overlap density ρ . The overlap density is itself the natural measure on the admissible configuration space: the Radon-Nikodym derivative of the transported measure with respect to the reference measure. Probability, in Phase Theory, is a feature of the geometry of Q , not an independent axiom about what measurements yield. This resolves one of the deepest and most persistently contested features of quantum foundations: what the Born rule means and why it holds. Within Phase Theory, it means that the probability of finding a system at configuration q is the density of phase overlap at that configuration, and it holds because that is the definition of probability in a theory whose fundamental object is phase overlap density.

Fourth: the clarification of Planck's constant. The reduced Planck constant $\hbar$ is, within Phase Theory, the ratio of the phase-curvature scale of the substrate Σ to the admissible ordering rate. It is a constant of nature — it does not vary from system to system — because the substrate is homogeneous at all scales currently accessible to physics: its phase-curvature scale and ordering rate are uniform across the regions of the substrate explored by laboratory and astronomical observations. Phase Theory predicts, however, that this homogeneity must break down near phase defects in the substrate — singularities in the phase field where the admissibility geometry is degenerate. Near such defects, the local phase-curvature scale and ordering rate may differ from their bulk values, and the effective $\hbar$ — the ratio of these two locally measured quantities — may vary. A spatially varying Planck

constant is a falsifiable prediction of Phase Theory, distinct from all prior approaches to quantum foundations. Its detection would constitute the most direct evidence for the substrate structure underlying Phase Theory [24, 29].

18. Conclusion

We have derived the Schrödinger equation as an effective dynamical law governing the transport of phase overlap density on the reduced descriptive space Q of Phase-Primary Quantum Theory, in the quantum-coherent regime defined by four explicit conditions: shallow admissibility basins, smooth phase gradients, negligible topological obstruction, and preserved coherence. The derivation requires no primitives beyond the admissibility geometry of Phase Theory. No wavefunctions are assumed — they emerge as overlap density amplitudes. No Hilbert space is assumed — it emerges as the space of square-integrable overlap densities. No Hamiltonian is postulated — it emerges as the generator of admissible phase flow scaled by the phase-curvature rate. No unitary axiom is assumed — unitarity emerges from the measure-preserving character of admissible flow in the QC regime. The imaginary unit is not fundamental — it encodes the 90-degree rotation between amplitude and phase-gradient transport under the symplectic structure of Q .

The Schrödinger equation is therefore a local chart on phase space: a compressed, extraordinarily accurate description of admissible overlap transport that is valid wherever the four QC conditions are met, and structurally limited wherever they fail. Four classes of predicted deviations — the nonlinear correction of order $|\alpha\kappa|/|\psi|^2\psi$, the non-unitary pruning term $-i\Gamma\psi$, the topological holonomy addition $\hat{A}\psi$, and the near-saturation anomalous decoherence — provide concrete experimental targets for probing the foundations of Phase Theory beyond the effective quantum level. None of these deviations is accessible to current experiments; all are in principle

accessible, and their eventual detection or definitive exclusion will constitute a decisive test of the theory.

The broader program of Phase Theory — to derive all of fundamental physics from a single phase-topological foundation — thus passes a critical internal consistency check: the most precisely tested equation in the history of science is recovered as a theorem within Phase Theory, not assumed as an axiom. The program continues with the derivation of relativistic quantum field theory (Paper 29), the emergence of gauge symmetries from phase topology (Paper 30), and the reconstruction of the gravitational field equations from substrate phase geometry (Paper 31).

The empirical success of quantum mechanics is not evidence against Phase Theory. It is evidence that the universe, at the scales and energies we have explored, overwhelmingly occupies configurations within the quantum-coherent regime. The task of physics is to find the boundary — to probe the limits of the QC approximation and discover the deeper structure that underlies it. That task has now been given a precise formulation.

“The Schrödinger equation is not the engine of reality. It is a local chart on phase space.”

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